MATH 180 Strategies of Problem Solving

W.R. Casper

Department of Mathematics
California State University Fullerton

August 19, 2025

Outline

1. SoPS Lecture 1
   - Welcome
   - New game
   - New game plus
   - Finishing up

Outline

What is this course about?

What is this course about?

Learn what mathematicians do

What is this course about?

Learn what mathematicians do

- play $\to$ discover patterns

What is this course about?

Learn what mathematicians do

- play $\rightarrow$ discover patterns
- abstract $\rightarrow$ express patterns mathematically

What is this course about?

Learn what mathematicians do

- play $\rightarrow$ discover patterns
- abstract $\rightarrow$ express patterns mathematically
- prove $\rightarrow$ justify why the pattern is true

What is this course about?

Learn what mathematicians do

- play $\rightarrow$ discover patterns
- abstract $\rightarrow$ express patterns mathematically
- prove $\rightarrow$ justify why the pattern is true

Glimpse advanced mathematics

What is this course about?

Learn what mathematicians do

- play $\to$ discover patterns
- abstract $\to$ express patterns mathematically
- prove $\to$ justify why the pattern is true

Glimpse advanced mathematics

- logic, set theory, cardinality

What is this course about?

Learn what mathematicians do

- play $\rightarrow$ discover patterns
- abstract $\rightarrow$ express patterns mathematically
- prove $\rightarrow$ justify why the pattern is true

Glimpse advanced mathematics

- logic, set theory, cardinality
- probability, combinatorics, graph theory

What is this course about?

Learn what mathematicians do

- play $\to$ discover patterns
- abstract $\to$ express patterns mathematically
- prove $\to$ justify why the pattern is true

Glimpse advanced mathematics

- logic, set theory, cardinality
- probability, combinatorics, graph theory

Make friends

What is this course about?

Learn what mathematicians do

- play → discover patterns
- abstract → express patterns mathematically
- prove → justify why the pattern is true

Glimpse advanced mathematics

- logic, set theory, cardinality
- probability, combinatorics, graph theory

Make friends

- math is social

What is this course about?

Learn what mathematicians do

- play $\to$ discover patterns
- abstract $\to$ express patterns mathematically
- prove $\to$ justify why the pattern is true

Glimpse advanced mathematics

- logic, set theory, cardinality
- probability, combinatorics, graph theory

Make friends

- math is social
- you'll see each other *a lot*

Course expectations

Course expectations

Grading

Course expectations

Grading

- pre-class assignments $\rightarrow$ 15%

Course expectations

Grading

- pre-class assignments $\rightarrow$ 15%
- in-class assignments $\rightarrow$ 25%

Course expectations

Grading

- pre-class assignments $\rightarrow$ 15%
- in-class assignments $\rightarrow$ 25%
- homework $\rightarrow$ 25%

Course expectations

Grading

- pre-class assignments $\rightarrow$ 15%
- in-class assignments $\rightarrow$ 25%
- homework $\rightarrow$ 25%
- seminars / research $\rightarrow$ 15%

Course expectations

Grading

- pre-class assignments $\rightarrow$ 15%
- in-class assignments $\rightarrow$ 25%
- homework $\rightarrow$ 25%
- seminars / research $\rightarrow$ 15%
- final exam $\rightarrow$ 20%

Course expectations

Course expectations

Daily Expectations

Course expectations

Daily Expectations

- complete pre-class assignments before class

Course expectations

Daily Expectations

- complete pre-class assignments before class
- attend every class

Course expectations

Daily Expectations

- complete pre-class assignments before class
- attend every class
- participate **actively**

Course expectations

Daily Expectations

- complete pre-class assignments before class
- attend every class
- participate **actively**
- solicit participation from group members

Course expectations

Daily Expectations

- complete pre-class assignments before class
- attend every class
- participate **actively**
- solicit participation from group members
- do your homework (or else)

Course expectations

Course expectations

Seminar Participation

Course expectations

Seminar Participation

- must attend at least three

Course expectations

Seminar Participation

- must attend at least three
- write a one-page report

Course expectations

Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

Course expectations

Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

Ongoing Seminars

Course expectations

Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

Ongoing Seminars

- pure math seminar

Course expectations

Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

Ongoing Seminars

- pure math seminar
- applied math seminar

Course expectations

Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

Ongoing Seminars

- pure math seminar
- applied math seminar
- math education seminar

Course expectations

Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

Ongoing Seminars

- pure math seminar
- applied math seminar
- math education seminar
- problem solving seminar

Course expectations

Seminar Participation

- must attend at least three
- write a one-page report
- your time at CSUF is what you make it!

Ongoing Seminars

- pure math seminar
- applied math seminar
- math education seminar
- problem solving seminar
- other events (workshops, colloquia, etc)

Course expectations

Course expectations

Research project

Course expectations

Research project

- what math research is

Course expectations

Research project

- what math research is
- do original research

Course expectations

Research project

- what math research is
- do original research
- write a final report

Course expectations

Research project

- what math research is
- do original research
- write a final report
- present results

Outline

1. SoPS Lecture 1
   - Welcome
   - New game
   - New game plus
   - Finishing up

Pinching Pennies

Pinching Pennies

Rules

Pinching Pennies

Rules

- arrange pennies in two piles

Pinching Pennies

Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*

Pinching Pennies

Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn

Pinching Pennies

Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny

Pinching Pennies

Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

Pinching Pennies

Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

Activity (20 mins)

Pinching Pennies

Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

Activity (20 mins)

- split into pairs

Pinching Pennies

Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

Activity (20 mins)

- split into pairs
- play 4-5 rounds

Pinching Pennies

Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

Activity (20 mins)

- split into pairs
- play 4-5 rounds
- record data: initial board, who won?

Pinching Pennies: Debrief

Pinching Pennies: Debrief

Let's combine the data ...

Pinching Pennies: Debrief

Let's combine the data ...

Question 1

is it better to be Player 1 or Player 2 ?

Pinching Pennies: Debrief

Let's combine the data ...

Question 1

is it better to be Player 1 or Player 2 ?

Question 2

if the piles are equal, then who should win?

Pinching Pennies: Debrief

Let's combine the data ...

Question 1

is it better to be Player 1 or Player 2 ?

Question 2

if the piles are equal, then who should win?

Fight me!

Vocabulary

Vocabulary

Optimal Play

Vocabulary

Optimal Play

A player is said to **play optimally** or to use an **optimal strategy** if their moves put them in the best possible final position, regardless of the moves of other players.

Vocabulary

Optimal Play

A player is said to **play optimally** or to use an **optimal strategy** if their moves put them in the best possible final position, regardless of the moves of other players.

Winning/Losing Position

Vocabulary

Optimal Play

A player is said to **play optimally** or to use an **optimal strategy** if their moves put them in the best possible final position, regardless of the moves of other players.

Winning/Losing Position

A **winning position** for a player in a game is a state of the game where if that player plays optimally, they are guaranteed to win. A **losing position** for a player is a game state where one of their opponents has a winning position.

Outline

1. SoPS Lecture 1
   - Welcome
   - New game
   - New game plus
   - Finishing up

Pinching Pennies – revisited

Pinching Pennies – revisited

Activity (20 minutes)

Pinching Pennies – revisited

Activity (20 minutes)

- arrange pennies in **three piles**

Pinching Pennies – revisited

Activity (20 minutes)

- arrange pennies in **three piles**
- which starts are losing positions for Player 1?

Pinching Pennies – revisited

Activity (20 minutes)

- arrange pennies in **three piles**
- which starts are losing positions for Player 1?

Submit your conjecture:

Break time!

Activity (5 minutes)

- get up, stretch get a cup of coffee
- chat up your neighbor, introduce yourself

Pinching Pennies – two equal piles

Pinching Pennies – two equal piles

Question 1:

Two piles with one penny each, who wins?

Pinching Pennies – two equal piles

Question 1:

Two piles with one penny each, who wins?

Question 2:

Two piles with two pennies each, who wins?

Pinching Pennies – two equal piles

Question 1:

Two piles with one penny each, who wins?

Question 2:

Two piles with two pennies each, who wins?

Question 3:

Two piles with three pennies each, who wins?

Pinching Pennies – two equal piles

Question 1:

Two piles with one penny each, who wins?

Question 2:

Two piles with two pennies each, who wins?

Question 3:

Two piles with three pennies each, who wins?

Copycat strategy

Player 2 copies what Player 1 does to the opposite pile

Our first proof

Our first proof

Theorem

Two equal piles of pennies is a losing position for Player 1.

Our first proof

Theorem

Two equal piles of pennies is a losing position for Player 1.

Proof

Our first proof

Theorem

Two equal piles of pennies is a losing position for Player 1.

Proof

Start with n pennies in each pile.

Our first proof

Theorem

Two equal piles of pennies is a losing position for Player 1.

Proof

Start with n pennies in each pile.
If Player 1 takes a whole pile, Player 2 wins.

Our first proof

Theorem

Two equal piles of pennies is a losing position for Player 1.

Proof

Start with n pennies in each pile.
If Player 1 takes a whole pile, Player 2 wins.
Otherwise Player 1 must reduce a pile to m pennies with $m < n$.

Our first proof

Theorem

Two equal piles of pennies is a losing position for Player 1.

Proof

Start with n pennies in each pile.
If Player 1 takes a whole pile, Player 2 wins.
Otherwise Player 1 must reduce a pile to m pennies with $m < n$.
Then Player 2 reduces the other pile to m pennies.

Our first proof

Theorem

Two equal piles of pennies is a losing position for Player 1.

Proof

Start with n pennies in each pile.
If Player 1 takes a whole pile, Player 2 wins.
Otherwise Player 1 must reduce a pile to m pennies with $m < n$.
Then Player 2 reduces the other pile to m pennies.
Now we're back where we started, but with less pennies!

Our first proof

Theorem

Two equal piles of pennies is a losing position for Player 1.

Proof

Start with n pennies in each pile.
If Player 1 takes a whole pile, Player 2 wins.
Otherwise Player 1 must reduce a pile to m pennies with $m < n$.
Then Player 2 reduces the other pile to m pennies.
Now we're back where we started, but with less pennies!
Repeat until Player 1 is forced to take a whole pile.

Outline

Question of the Day

Puzzle

You have 15 coins. On each turn you may remove 1, 2, or 3 coins. The last player to move wins. Who has a winning strategy?

Final thoughts

Reflection

- What did you notice about how mathematicians work compared to how you expected?
- What differences do you see between experimentation (play) and proof?
- How is this class different from the math you've done before? What do you think is the key to your success in this class?

Submit your responses by **Tonight!**